

Scent-Based Landmark and Communication for EM Hostile Environments

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Abstract. An effective use of mobile robots is their ability to traverse environments which can be hostile or dangerous to human life. This problem is further accentuated with environments where communications using electromagnetic wave may prove ineffective, such as caves, or collapsed infrastructure. This paper proposes a method of establishing a simple communication between robots using chemicals. The communication relies on the pathfinder robot leaving scent packages to be picked up by another robot which can interpret the action. In addition a simple package structure is proposed along with preliminary information capacity of communication is analysed. The results show that if proper care is given on the selection of chemicals, temporary landmarks could be established for robots to communicate without effecting the surrounding environment.

Keywords: navigation, communication, mobile robotics, molecular communications

1 Introduction

Electro-Magnetic (EM) wave communication has changed how information is exchanged and allowed communication of immeasurable distances to be measured in milliseconds. It also allows us to communicate with robots which can act in a wide variety of environments and has shown to be efficient and robust for most cases [1]. However, not every environment can easily be navigated with EM-waves as caves and places which encountered natural disasters can prove hard to transmit information [2]. The primary problem in using EM-waves is that they are attenuated as they pass through a conducting medium. There are numerous detriments caused by using EM-waves such as; range-dependent effects [3], depolarisation effects [4], antenna height gain effects [4], the subterranean gallery geometry [5], and multi-scattering effect [6] on the propagating radio waves in caves.

An important application of interest is disaster scenarios [7], where communication systems can fail and environment can become inhospitable for human exploration [1,8,9]. While temporary infrastructure for communication can quickly be established, these can't necessarily penetrate the rubble of collapsed buildings

and can be generally the limiting factor of their applicability [10] so algorithms for multi-robot exploration in these situations must be able to accommodate the limited and variable communication available to them [11].

In a disaster scenario, It is necessary for the robots to complete exploration as quickly and efficiently as possible, and minimising repeated coverage of any area in the initial exploration phase helps reduce exploration time. With a small team of robots, it is common to use beacons to mark explored areas [12–14], inspired by animal behaviour [15], where these beacons enable the robots to reduce repeated coverage and explore larger environments without having to create maps.

Regarding a viable alternative to the aforementioned challenges is the use of scent based communication which has significant research behind it [16–21] along with Molecular Communication (MC), where information is encoded into a chemical source [22], which this research attempts to implement.

In this paper an alternative framework for mobile robotic communication is proposed for use in cave exploration/disaster scenarios for rapid exploration of the environment. To achieve this, scent-based packages will be used which will be placed by robots exploring, to provide simple transmission of information where these chemical messages will be designed using volatile organic compounds. To achieve proper data transmission, concepts developed in MCs will be used to analytically study the Bit-Error Rate (BER) and Mutual Information (MI) and its reliability of data acquisition.

2 Defining the Physical Layer

2.1 A Mathematical Model

To understand how the landmarks behave, a mathematical model is required. To model the medium, the following assumptions were made to derive an analytical model: the environment has no advection (i.e., $\mathbf{v} = 0$), The space in xy -axis have no boundaries, The system has no sink/source (i.e., $K = 0$), The detector sensor has perfect absorption, and The diffusion constant. Under these conditions, the landmarks can be modeled using the diffusion equation [23]:

$$c_t = D(c_{xx} + c_{yy} + c_{zz}) + K,$$

where c is the concentration of the mass (kg m^{-3}), t is the mass transfer duration (s), D is the diffusion coefficient ($\text{cm}^2 \text{s}^{-1}$), and K is the sink and/or the source. The solution for this equation can be derived using the following initial boundary conditions, known as the “thin-film solution” in the literature [23]:

$$\begin{aligned} c(|x| > 0, |y| > 0, z > 0, t_0) &= 0, \\ c(x = 0, y = 0, z = 0, t_0) &= M_0 \delta(x)\delta(y)\delta(z), \\ c(|x| \rightarrow \infty, |y| \rightarrow \infty, |z| \rightarrow \infty, t) &= 0, \end{aligned}$$

where M_0 is the initial mass injected into the environment (kg), x is the Cartesian propagation dimension, t_0 is the initial time (s) and $\delta(x)$ is the one-dimensional

Dirac delta function. Under these conditions, the solution with an unbounded domain in x, y and a boundary at $z = 0$, can be expressed as follows:

$$C(x, y, z, t) = \frac{M}{\sqrt{(4\pi D_x D_y D_z t)^3}} \exp\left(-\frac{x^2}{4D_x t} - \frac{y^2}{4D_y t} - \frac{z^2}{4D_z t}\right) \quad (1)$$

The diffusion coefficient (D), based on environment can be estimated as [24];

$$D = \frac{2}{3} \sqrt{\frac{k_B^3 T^3}{\pi^3}} \sqrt{\frac{1}{2m_A} + \frac{1}{2m_B}} \frac{4}{P(d_A + d_B)^2},$$

where k_B is the Boltzmann constant¹, P is pressure, T is temperature, m_A, m_B and d_A, d_B are the molecular masses and diameters of chemical A and B respectively. Eq. (1) quantifies the concentration value of the sample in a given time (t) and 3D space (x, y, z). The measured mass in a given detection (θ) can be calculated by integrating the concentration function in the desired volume:

$$\theta = \int C \, dx \, dy \, dz.$$

As mentioned, the system has no sink/source and therefore, the chemicals used in the transmission can either be in motion (θ_T) or have been absorbed by the detector (θ_A). Therefore, both mass values must add up to the initial mass in the beginning of the transmission.

$$M = \theta_T + \theta_A, \quad \text{therefore} \quad \theta_A = M - \theta_T.$$

By integrating the concentration function with respect to space, the particles present in the environment can be calculated. To calculate chemicals absorbed by the detector, the integration function is subtracted from the injected mass [25–27]:

$$\theta(x, y, z, t) = M - \int_{-x_\epsilon, -y_\epsilon, 0}^{+x_d, y_d, z_d} C(x, y, z, t) \, dx \, dy \, dz. \quad (2)$$

where subscript d defines distance from the detector to the origin point and ϵ is the distance particles travel against the flow². The solution to the integration given in Eq. (2) for transmission with boundaries is given in Eq. (3).

$$\begin{aligned} \theta_1(x_d, x_\epsilon, y_d, y_\epsilon, t) &= M - \frac{M}{4} \left[\operatorname{erf}\left(\frac{x_d}{\sqrt{4Dt}}\right) + \operatorname{erf}\left(\frac{x_\epsilon}{\sqrt{4Dt}}\right) \right] \\ &\times \left[\operatorname{erf}\left(\frac{y_d}{\sqrt{4Dt}}\right) + \operatorname{erf}\left(\frac{y_\epsilon}{\sqrt{4Dt}}\right) \right] \left[\operatorname{erf}\left(\frac{z_d}{\sqrt{4Dt}}\right) \right] \end{aligned} \quad (3)$$

where $\operatorname{erf}(x) = 2/\sqrt{\pi} \int_0^x e^{-t^2} dt$. The chemicals absorbed by the detector (θ_1) in a given period of T is given below:

$$M_R = \theta_1(x_d, t = T) - \theta_1(x_d, t = 0).$$

Therefore, the removal of chemicals from the detector (θ_0) to the outside environment can be expressed by the following expression.

$$\begin{aligned} \theta_0(x_d, x_\epsilon, y_d, y_\epsilon, t) &= \frac{M_R}{4} \left[\operatorname{erf} \left(\frac{x_d}{\sqrt{4D_x t}} \right) + \operatorname{erf} \left(\frac{x_\epsilon}{\sqrt{4D_x t}} \right) \right] \\ &\times \left[\operatorname{erf} \left(\frac{y_d}{\sqrt{4D_y t}} \right) + \operatorname{erf} \left(\frac{y_\epsilon}{\sqrt{4D_y t}} \right) \right] \operatorname{erf} \left(\frac{z_d}{\sqrt{4Dt}} \right). \end{aligned} \quad (4)$$

As can be seen from Eq. (3) and Eq. (4), the mass parameter is different in each equation: former being the mass injected into the environment (M) and the latter is the mass absorbed by the detector (M_R). The application and the process of introduction/removal of particles in 1D can be seen in detail in [26, 27].

Environmental Noise To properly model the limits on information, and understanding of the environmental noise is required. Based on experiments done in [26] it was determined to be an Additive White Gaussian Noise (AWGN) and therefore will be used in the modelling:

$$\text{AWGN} = \mathcal{N}(\mu, \sigma^2).$$

3 Defining the Landmark

3.1 Operation Principle

A significant bottleneck of communicating in an EM adversarial environment is the communication between robots themselves in which this paper proposes an alternative method. A key problem is the passing of information from one robot to another to give information deemed important. A simple example would be the its usage in a fork in a cave during initial exploration, which the diagram can be seen in Fig. 1. In this scenario, robot A (R_A) would go in the left fork, however

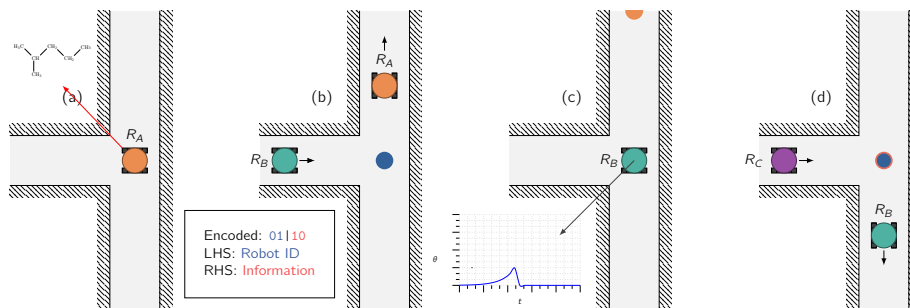


Fig. 1: A diagram describing the scent-based land-marking. (a) When a robot A (R_A) enters a cross-road, it sprays the floor with a compound which contains information about its identity and which path it took (b) Robot B (R_B) enters and enters the same cross-roads R_A did but detect a chemical. (c) (R_B) read the chemical signal and determines which way R_A went and decides to pick a different route and adds this information to the chemical pool. (d) (R_C) approaches.

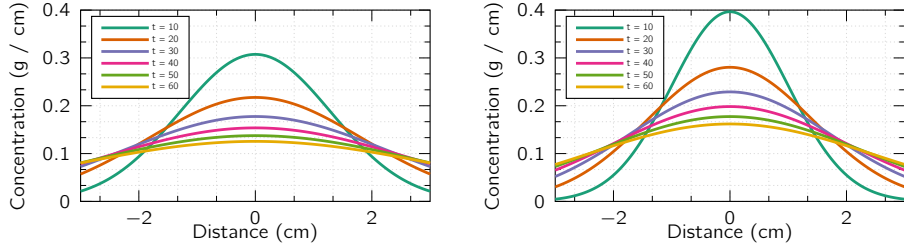


Fig. 2: Comparison of two organic compounds concentration (a) Pentane ($8.42 \times 10^{-2} \text{ cm s}^{-2}$) (b) n-Octyl alcohol ($5.06 \times 10^{-2} \text{ cm s}^{-2}$).

for efficiency, the robot B (R_B) following R_A needs to know to not repeat what R_A did. This can be accomplished by R_A introducing a scent in a specific part on the path where it can communicate with robot R_B as can be seen in Fig. 1.

It goes without saying, the information carrying speed is nowhere near to that of EM, however, the goal here is not to have a high-throughput bi-directional communication but to leave short messages for other robots to find.

3.2 Volatile Organic Compounds

An important factor to allow information to be embedded to a surface and to retain the content of the information is to use chemicals which can generate detectable vapour for a practically useful amount of time. An to that benefit, a useful class of chemical compound is Volatile Organic Compound (VOC) [29]. These compounds have high vapour pressure at room temperature and are relatively common-place. These chemicals also allow communication between animals and plants, such as attracting pollinators, or protection from predators [30–32] which makes it a good candidate for robotic communications using bio-mimicry.

When a robot sprays a compound onto a surface, the open surface of the liquid will enable evaporation due to the imbalance of thermodynamic equilibrium. The evaporation is defined not only by geometrical parameters such as thickness and surface area but also by its temperature and chemical composition. However, for practical applications, simplifications can be made [26]. As discussed in Section 2, the primary agent which defines the propagation is diffusion, and therefore that parameter will be looked to observe the range of velocity. A selection of diffusion

Table 1: Diffusion coefficients of selected organic compounds. The diffusion values are for air [28], measured in $\text{cm}^2 \text{ s}^{-1}$.

Compounds Diffusion		Compounds Diffusion	
Cholorobenzene	8.8×10^{-2}	Toluene	8.7×10^{-2}
Ethylbenzene	7.5×10^{-2}	o-Xylene	8.7×10^{-2}

coefficient values are shown in Table 1 and the behaviour of two chemicals with different diffusion values is shown in Fig. 2.

The diffusion coefficients can also be calculated using the following empirical relation [33] for simulation purposes:

$$I_D = 9.490 - \sqrt{90.136 - 4 \left(\frac{kT}{\epsilon_{AB}} \right)^2 - 4.38 \left(\frac{kT}{\epsilon_{AB}} \right) + 9.6398} \quad (5)$$

where T can range from 273 K to 673 K, which is a comfortable range for a robot to operate on and ϵ_{AB} is the energy of molecular interaction, which for air the value is taken to be 9.7 [34].

3.3 Detection of Chemical Odour

To measure the chemical concentration with the accuracy needed for embedding information, a chemical sensor capable of distinguishing scents is needed. To that end there are two options commercially available:

Electronic Nose A standard tool which is used in detecting and analysing odours and providing precise information, known as a “fingerprint” about mixed gases and scents [35] with studies done to implement this technology to mobile robotics [36].

Mass Spectrometers A powerful analytic technique used to quantify known and unknown compounds within a sample, and determine the chemical properties of different molecules, which has been use by robots as a mobile lab instead of a sensor for the robot itself [37, 38].

3.4 Chemical Quadrature Amplitude Modulation

While the ability to put information is useful, it is nevertheless in our advantage to squeeze as much information as possible and to that end a method using multiple chemicals needs to be employed.

As the proposed mode of communication requires two-axis (i.e., multi-chemical communication) it could be useful to separate the communication channel to two independent channels. While this increases the complexity of the communication, there are advantages compared to EM-based communication. An important one worth mentioning is that if adequately separate chemicals are chosen, there is no possibility of cross-talk and as communication is done through particles instead of waves, there are no problems with scattering or reflection.

The amount of mappable item onto a chemical amplitude can be expressed by the cardinality of the input set with following equation [39]:

$$\text{card}(\text{ChQAM}(k, m)) = k^m, \quad (6)$$

where k is the number of modulated chemical levels and m is the number of distinguishable chemicals used in the transmission of information. Based on this,

the channel capacity of an n block transmission with memory effects can be written using the following equation [40]:

$$C \triangleq \lim_{n \rightarrow \infty} \sup \frac{1}{n} I(X^n; Y^n). \quad (7)$$

In Eq. (7), $I(X^n; Y^n)$ represents the mutual information between the input X and the output Y alphabet with the following relation [41]:

$$I(X^n; Y^n) = H(Y^n) - H(Y^n|X^n), \quad (8)$$

where $H(Y^n)$ is the Shannon entropy of the probability vector Y and $H(Y^n|X^n)$ is the conditional entropy of Y given X .

$$H(X^n) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i),$$

$$H(Y^n|X^n) = - \sum_{x \in \mathcal{X}} p(x) \sum_{y \in \mathcal{Y}} p(y|x) \log_2 p(y|x).$$

In macro-scale MC, the effects of Inter-Symbol Interference (ISI) on the channel behaviour needs discussion. MC uses particles to conduct transmission and as the transmission evolves the leftover chemicals in the environment and/or the detector may pose a hindrance on the usable information rate. This can cause incorrect decoding at a different rate (i.e., $p(0|1) \neq p(1|0)$). Therefore, every value in the probability matrix needs to be calculated individually as the ISI may effect the values unevenly making the behaviour of the channel asymmetric [42].

As indicated, a unique attribute of MC is the ability to increase the amount of carrier signals in a given channel. For example, in [39] it was shown transmitting multiple chemicals at the same time is possible. In addition, this doesn't limit the number of concurrently usable chemicals to only two as more chemicals can be used simultaneously without cross-talk³. These chemicals have the possibility of transmitting information with negligible interference when chosen carefully⁴.

In the analysis, two chemicals (m_I , m_Q) were chosen for the application. It is assumed the messenger chemicals have no alignment in their mass spectrum which can cause interference. The number of modulated levels (k) will be $k = \{2\}$ which a sample of the received signal can be seen in Fig. 3. To theoretically define the Symbol-Error Rate (SER) and to validate the proposed model, multivariate normal distribution is employed. The probability density function (PDF) of the distribution is given as;

$$f(\mathbf{x}; \mu, \Sigma) = \frac{1}{\sqrt{(2\pi)^d |\Sigma|}} \times \exp \left(-\frac{1}{2} (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) \right),$$

where d is the number of dimensions, μ is the expected value of $\mathbf{x}$ (i.e., $\mu = E\mathbf{x}$) and Σ is the covariance matrix with the following identity,

$$\Sigma = \text{cov}[\mathbf{x}_i, \mathbf{x}_j] = E[(\mathbf{x}_i - E[\mathbf{x}_i])(\mathbf{x}_j - E[\mathbf{x}_j])],$$

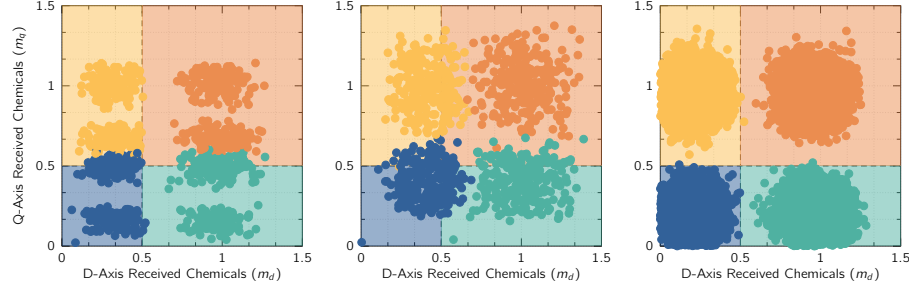


Fig. 3: The detected concentration levels of Chemical Quadrature Modulation (ChQAM) with a SNR of 20 dB. From left to right (a) With a diffusion ratio of 3 (b) distance of 1cm (c) Listening rate of 30 s. The parameters of the simulations are shown in Table 2.

and $|\Sigma|$ is the determinant of Σ . To calculate the probability occurring during transmission, the conditional probability of the distribution is applied [43].

$$\mathbf{P} \begin{bmatrix} a_x \leq x \leq b_x \\ a_y \leq y \leq b_y \end{bmatrix} = \int_{a_x}^{b_x} \int_{a_y}^{b_y} f_{X,Y}(x, y) dy dx. \quad (9)$$

4 Communication Tests

To understand the feasibility of the detection of chemicals, a bit-error-rate and mutual information study was conducted. While a scent landmark can't generate a stream of bits similar to a transmitter, nevertheless it is an information source and therefore a BER analysis would provide with the error probability of information or how much information can be embedded using a scent landmark using ChQAM. The simulation results can be seen in Fig. 4 with detected values in Fig. 3. The parameters of simulation are given in Table 2.

Table 2: Simulation Parameters for the communication study. Mass is measured in g, diffusion is measured in $\text{cm}^2 \text{s}^{-1}$.

Parameters	Diffusion	Distance	Symbol
Seed	42 and 89	42 and 89	42 and 89
Mass	1	1	1
Levels	2	2	2
Bit Counts	20.000	20.000	20.000
Diffusion	[0.5, 1, 1.5, 2.0, 2.5]	8.8×10^{-2}	8.8×10^{-2}
Distance	1	[0.5, 1, 1.5, 2.0, 2.5]	1
Time	30	30	[30, 60, 90, 120, 150]

In the study, three primary parameters were tested: diffusion coefficient, distance, and listening time (i.e., symbol time). In the diffusion study the value

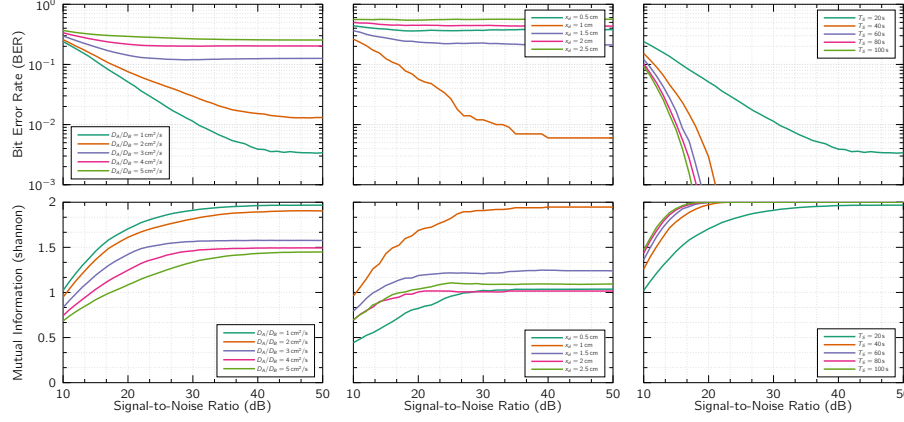


Fig. 4: Simulation results of scent-based detection system. From top-left first row; signal-to-noise v. bit-error rate ratio comparison of (a) different diffusion coefficient ratios, (b) detection distance (c) listening duration. From left second row; (d) signal-to-noise ratio v. mutual information of (a) different diffusion coefficient ratios, (b) detection distance (c) listening duration.

is kept at 8.8×10^{-2} with each iteration the ratio of the was changed. The simulation range for Signal-to-Noise Ratio (SNR) was kept from 10 to 50 dB.

As can be seen, the diffusion ratio (D_A/D_B) plays an important roles in the performance of mutual information or the error rate of detection, whereas distance and symbol time play a significantly more prominent role as to be expected. From the perspective of diffusion, this implies the correct decision of diffusion constant for the landmarks needs to be explicit as slight variations on the diffusion parameters can presents an asymmetrical detection of bits which can cause additional detection of errors which can also be seen in Fig. 3, sub-figure (a). As can be seen in the figure, the increase of the diffusion coefficient, while increases the propagation speed, also increases the dispersion of the chemical and causes the signal to degrade, which in the end causes the signal to be wrongly classified.

In regards to the symbol length, or the duration in which signal is being read, the improvements are significant as more time is devoted to reading the signal. Increasing the signal reading time from 30 to 60 s increases the readable information content from 1.5 to 1.8 shannon with SNR being 15 dB. This shows that, under less than ideal environments for chemical communication, a longer period of listening can produce higher information content to be read accurately. The SNR shows significant rate of improvement up to 30 dB with additional improvement gives incremental gain.

5 Conclusion and Future Work

In this work a proof-of-concept method was shown which implemented chemicals for land-marking instead of using vision or EM waves. In pursuit of this

idea, a mathematical framework was developed based on the diffusion equation. Constraints were introduced to aid in its simplification and a model was derived which quantifies the signal received by a sensor. Using the model, channel calculations were conducted to see the validity of using scent land-marking to convey information. The results have shown that when chemicals with suitable values (i.e., low diffusivity) where both chemical having similar diffusivity coefficient were chosen, it is possible for a mobile robot to detect the scent embedded to the environment and decode information about the previous robot which left the message minutes later.

Future-work will focus on increasing the information embedded to chemicals using different encoding methods and conduct field experiments.

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¹ $k_B = 1.380\,609 \times 10^{-23} \text{ J K}^{-1}$

² The ϵ term is there for mathematical bookkeeping as, practically it can be ignored without any loss in accuracy.

³ This statement is true as long as proper care is taken on the chemicals used.

⁴ i.e., when chemicals have no alignment in their mass spectrum.

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